Automated Experiment Report

Run folder: run_20260419_203951

Prepared on 2026-04-19 16:48:37. This report was created automatically from saved run artifacts using deterministic Python rules only.

Model family Direct Hurdle Tabular Compact Feature Recipe	**Horizons** 24, 48
Platform Linux-6.6.113+-x86_64-with-glibc2.35	**XGBoost device** cpu
Classifier configs 2	**Regressor configs** 2
Pairs per horizon 4	**Total fold fits** 8

Executive summary

- The run archived 143 tracked artifacts.
- Best 24h pair: h24__clf_xgbc01__reg_xgbr01.
- Best 48h pair: h48__clf_xgbc01__reg_xgbr01.
- For horizon 24h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 96.7632; Improved by 3.5422 (3.53%). Stage-1 ROC AUC was 0.8744, indicating strong discrimination.
- For horizon 48h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 72.7500; Improved by 2.6061 (3.46%). Stage-1 ROC AUC was 0.8661, indicating strong discrimination.
- Submission-style CSVs were exported from: Best validation candidate.
- Demo-only evaluation files are present. They are useful for sanity checks only and are not the hidden challenge evaluation.

Configuration and modeling decisions

- Architecture: the run uses a two-stage direct hurdle pipeline ('direct_hurdle_tabular_compact_feature_recipe'). Stage 1 predicts whether outage is non-zero; Stage 2 predicts outage size conditional on positive outcomes only.
- Combination rule: the final expected outage equals stage-1 non-zero probability times the stage-2 positive prediction.
- Target transform: Stage 2 uses log1p on positive outages before fitting the regressor.
- Forecast horizons: 24h, 48h.
- Forecast-origin sampling: every eligible hour was used.
- Minimum training history before the first fold: 21 days (504 h).
- Maximum training history: capped at 28 days (672 h).

- Temporal validation design: manual aligned splits were specified - 24h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-11 23:00:00; 48h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-11 23:00:00.
- Target lag features: 1, 24 hours.
- Weather screening: up to 10 weather variables were allowed; 10 were retained in this run.
- Weather outlier handling: enabled with 'clip_to_nan_then_linear_interpolate', so extreme weather values were clipped rather than turned into missing values.
- Execution hardware: GPU acceleration was not available; the run executed on CPU.
- Prediction export policy: submission CSVs were generated from Best validation candidate.

Search space summary

component	families	config_count
Classifier	logistic_enet, xgb_classifier	2
Regressor	elasticnet_regression, xgb_regressor	2

Run inventory

The table below shows the top-level artifact folders detected in the run. The checkpoints folder is runtime infrastructure and may be intentionally absent from shared GitHub uploads.

subfolder	present	file_count	note
tables	Yes	34	
predictions	Yes	12	
models	Yes	40	
metrics	Yes	28	
plots	Yes	17	
checkpoints	Yes	10	Runtime-only resume artifacts; may be omitted from shared GitHub uploads.

Temporal evaluation design

The backtest table below maps the training and validation windows used for each horizon and fold.

horizon	fold_id	val_origin_idx	val_origin_timestamp	train_origin_start_idx	train_origin_start_timestamp	train_origin_max_idx	train_origin_max_timestamp	validation_window_start	validation_window_end
24	1	1,727	2023-06-11 23:00:00	1,464	2023-06-01	1,703	2023-06-10 23:00:00	2023-06-12	2023-06-12 23:00:00
48	1	1,727	2023-06-11 23:00:00	1,464	2023-06-01	1,679	2023-06-09 23:00:00	2023-06-12	2023-06-13 23:00:00

Feature frame and data overview

metric	value
rows_in_model_df	179,363
base_numeric_features	17
categorical_features	0.0000
feature_warmup_hours	24
max_horizon	48
selected_weather_feature_count	10

metric	value
mean_outage	45.2483
median_outage	0.0000
std_outage	452.2684
max_outage	23,346
zero_rate	0.7047
nonzero_rate	0.2953

Selected weather features retained in this run: 10.

feature	abs_corr_with_outage	signed_corr_with_outage
vstm	0.0539	-0.0539
unknown_2	0.0523	0.0523
veg	0.0522	0.0522
vvcsh	0.0488	-0.0488
unknown	0.0439	0.0439
v	0.0422	-0.0422
ustm	0.0415	-0.0415
mdens	0.0408	0.0408
vucsh	0.0365	-0.0365
lai	0.0350	0.0350

Candidate ranking by horizon

Rows are ranked within each horizon using average county RMSE expected first and overall RMSE expected second.

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
1	24	xgb_classifier	xgb_regressor	96.7632	278.1097	3.5422	14.4496	0.8744	450.8753	h24__clf_xgbc01__reg_xgbr01
2	24	logistic_enet	xgb_regressor	97.3928	279.4119	2.9125	13.1473	0.6743	450.8753	h24__clf_logen01__reg_xgbr01
3	24	xgb_classifier	elasticnet_regression	99.6481	289.8708	0.6573	2.6885	0.8744	476.6630	h24__clf_xgbc01__reg_enet01
4	24	logistic_enet	elasticnet_regression	100.4998	289.5774	-0.1945	2.9819	0.6743	476.6630	h24__clf_logen01__reg_enet01
1	48	xgb_classifier	xgb_regressor	72.7500	200.1073	2.6061	9.8176	0.8661	340.8096	h48__clf_xgbc01__reg_xgbr01
2	48	logistic_enet	xgb_regressor	73.5494	202.0661	1.8067	7.8588	0.6692	340.8096	h48__clf_logen01__reg_xgbr01

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
3	48	xgb_classifier	elasticnet_regression	74.8222	208.0768	0.5339	1.8481	0.8661	357.3121	h48__clf_xgbc01__reg_enet01
4	48	logistic_enet	elasticnet_regression	75.7235	208.2349	-0.3674	1.6900	0.6692	357.3121	h48__clf_logen01__reg_enet01

Winning pairs by horizon

horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage1_f1	mean_stage2_positive_rmse	pair_tag
24	xgb_classifier	xgb_regressor	96.7632	278.1097	3.5422	14.4496	0.8744	0.6631	450.8753	h24__clf_xgbc01__reg_xgbr01
48	xgb_classifier	xgb_regressor	72.7500	200.1073	2.6061	9.8176	0.8661	0.6325	340.8096	h48__clf_xgbc01__reg_xgbr01

Horizon 24h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 96.7632 versus a zero baseline of 100.3053. Improved by 3.5422 (3.53%).

Horizon 48h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 72.7500 versus a zero baseline of 75.3561. Improved by 2.6061 (3.46%).

Stage-specific summaries

Stage 1 classifiers

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
24	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8744	0.8496	0.1485	0.6631
24	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.6743	0.5620	0.2278	0.5554

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
48	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8661	0.8067	0.1389	0.6325
48	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.6692	0.5348	0.2234	0.5226

Stage 2 regressors

horizon	regressor_name	regressor_params	mean_stage2_positive_rmse	mean_stage2_positive_mae	mean_stage2_positive_rows_train
24	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	450.8753	167.2319	140,227
24	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	476.6630	177.8664	140,227
48	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	340.8096	105.0261	251,115
48	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	357.3121	114.1107	251,115

Submission and refit artifacts

Submission model summary

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
24	Best validation candidate	h24__clf_xgbc01__reg_xgbr01	1	xgb_classifier	xgb_regressor	96.7632	278.1097

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
48	Best validation candidate	h48__clf_xgbc01__reg_xgbr01	1	xgb_classifier	xgb_regressor	72.7500	200.1073

Final refit summary

horizon	bundle_source	pair_tag	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected
24	Final refit	h24__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	96.7632	278.1097
48	Final refit	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	72.7500	200.1073

Prediction exports

file	rows	prediction_column	min	max	mean	std
two_stage_pred_24h.csv	1,992	pred	0.4018	326.1926	7.4901	26.0763
two_stage_pred_48h.csv	3,984	pred	0.2981	250.4897	6.5934	20.7872
final_refit_pred_24h.csv	1,992	pred	0.0094	543.6840	10.1087	45.8530
final_refit_pred_48h.csv	3,984	pred	0.0053	364.3990	6.6188	28.0733

The predictions folder also contains 8 pair_*_val_predictions.csv file(s) with row-level validation outputs for candidate analysis.

Demo-only evaluation files

These metrics were computed against any demo labels bundled with the repository. They are useful for quick checks, but they are not the official hidden-test challenge results.

file	overall_rmse	overall_mae	avg_county_rmse	n_rows	n_locations
demo_eval_24h.json	26.9925	6.9980	7.3891	1,992	83
demo_eval_48h.json	21.6687	6.1163	6.4653	3,984	83
demo_eval_final_refit_24h.json	46.8373	9.8083	10.6730	1,992	83
demo_eval_final_refit_48h.json	28.7380	6.3527	6.9581	3,984	83

Reusable model artifacts

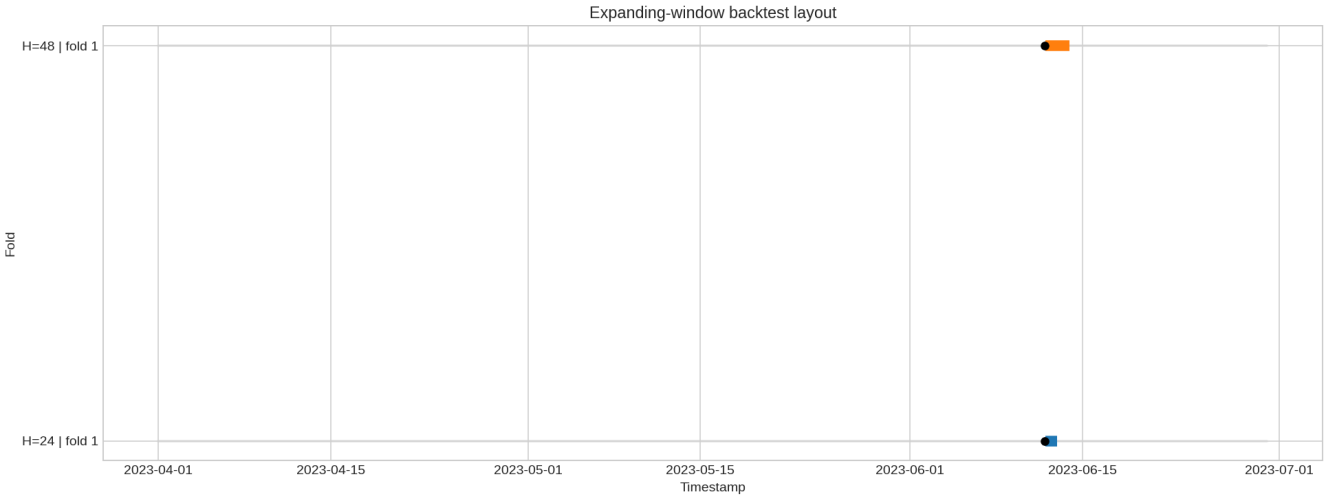
horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
24	1	h24__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	pair_h24__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib	pair_h24__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h24__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib

horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
48	1	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	pair_h48__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib

Visual appendix

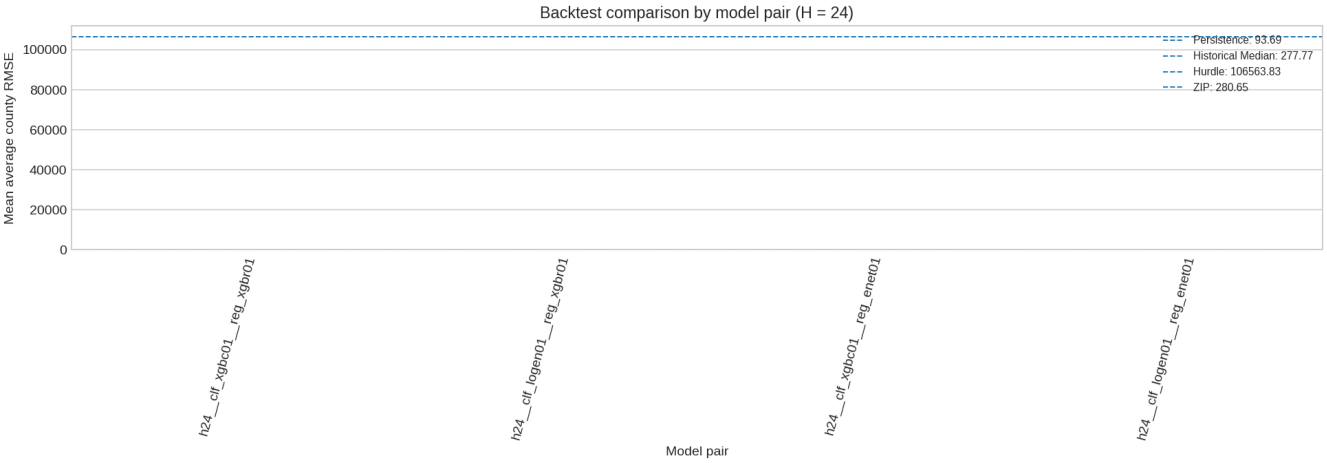
The following figures were read directly from the run's plots folder and inserted without GenAI interpretation.

Backtest Layout



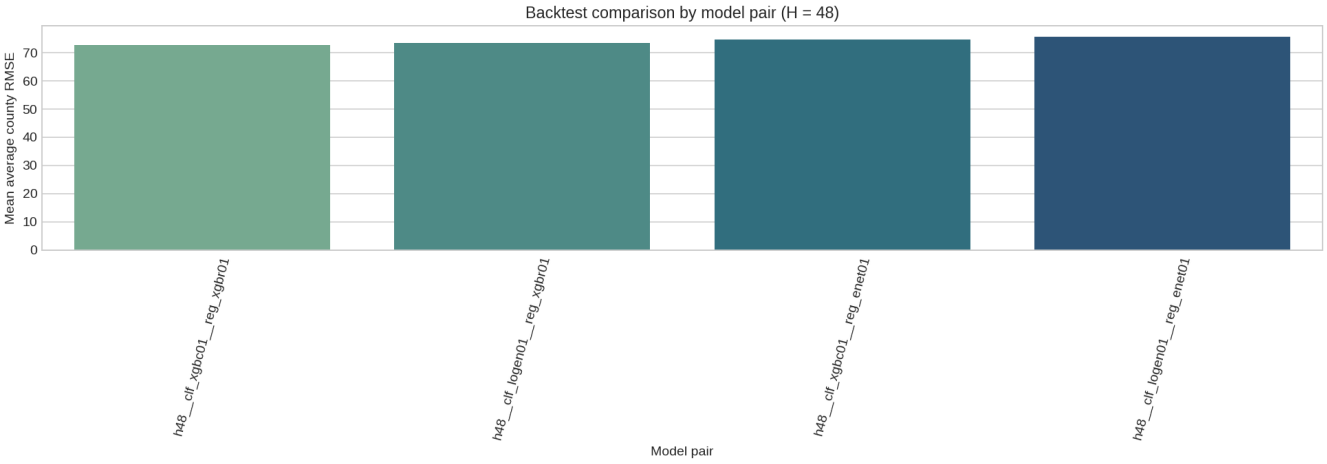
backtest_layout.png

Pair Comparison 24H



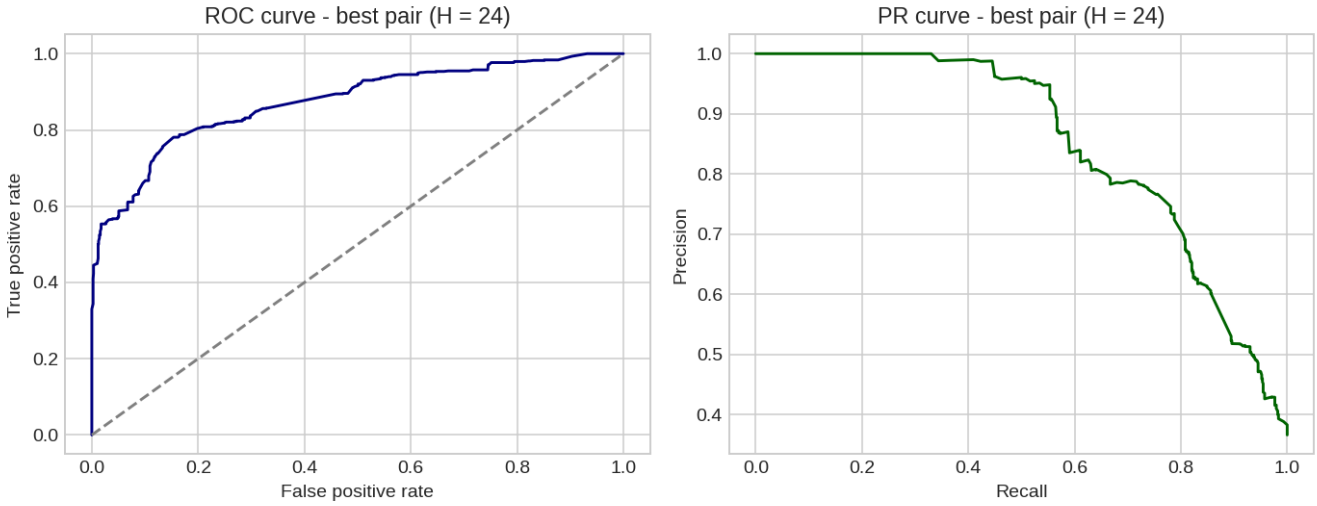
pair_comparison_h24.png

Pair Comparison 48H



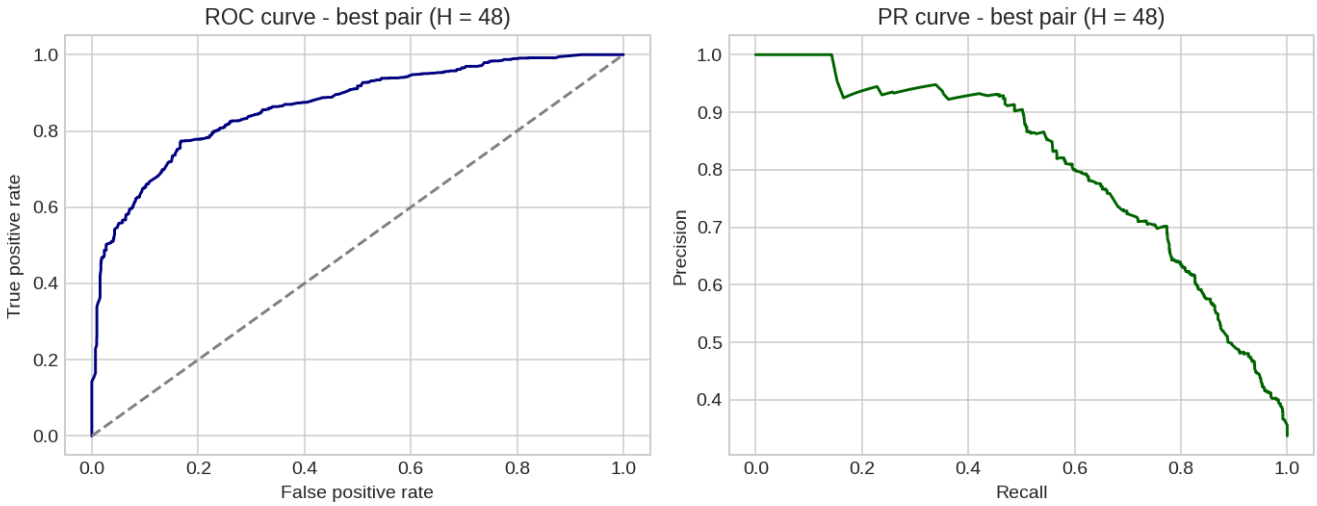
pair_comparison_h48.png

Best Pair Curves 24H



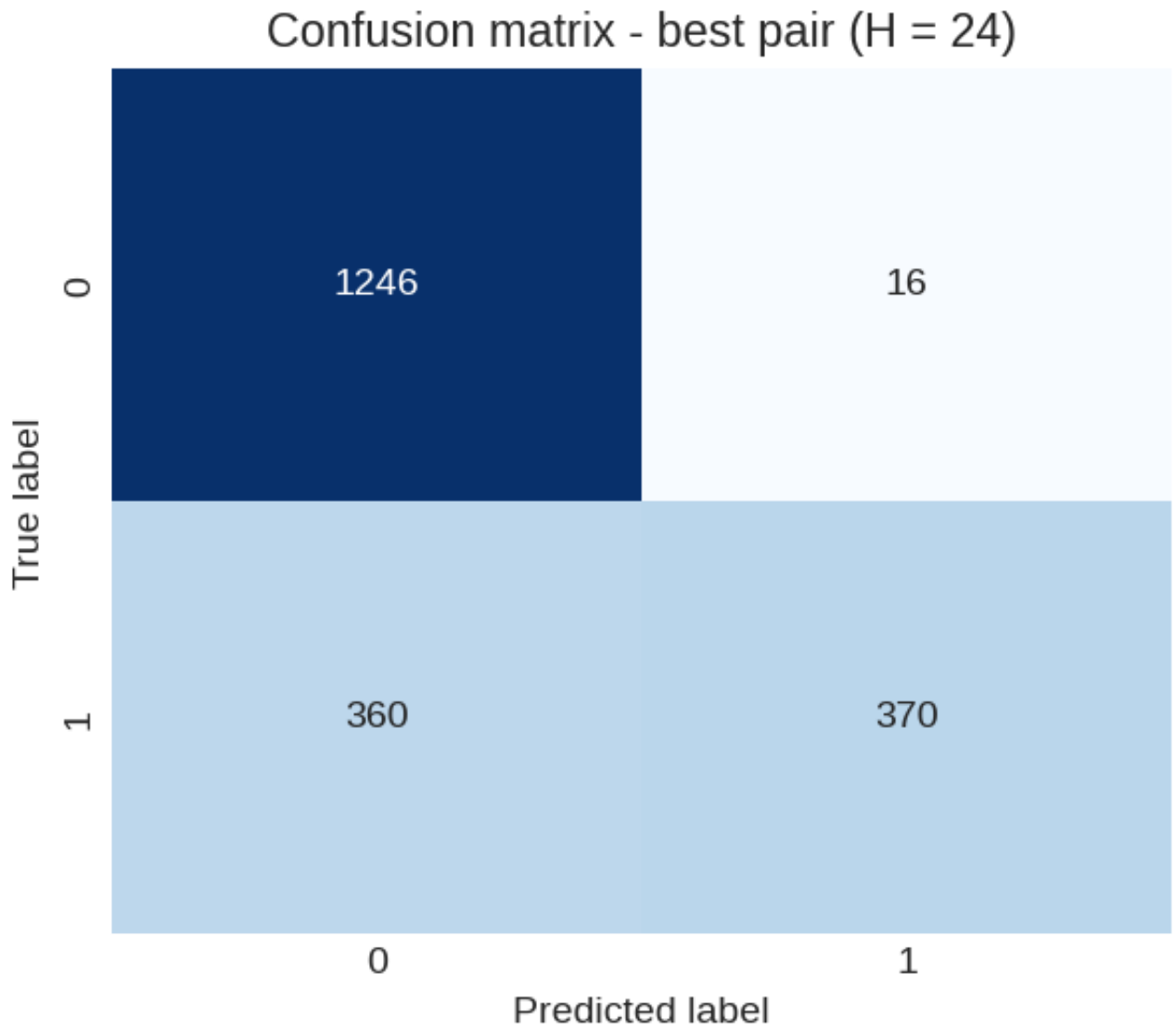
best_pair_curves_h24.png

Best Pair Curves 48H



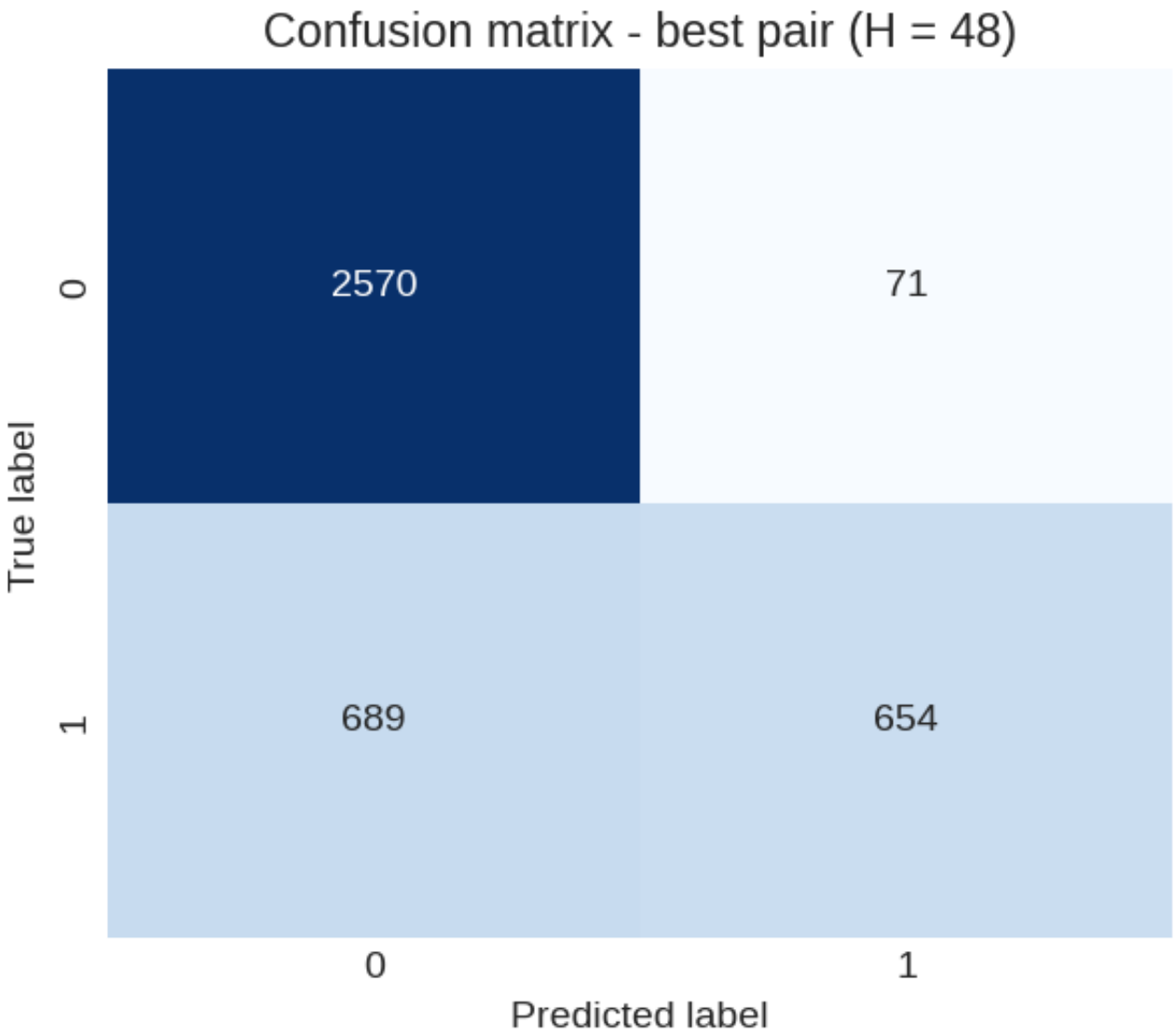
best_pair_curves_h48.png

Best Pair Confusion 24H



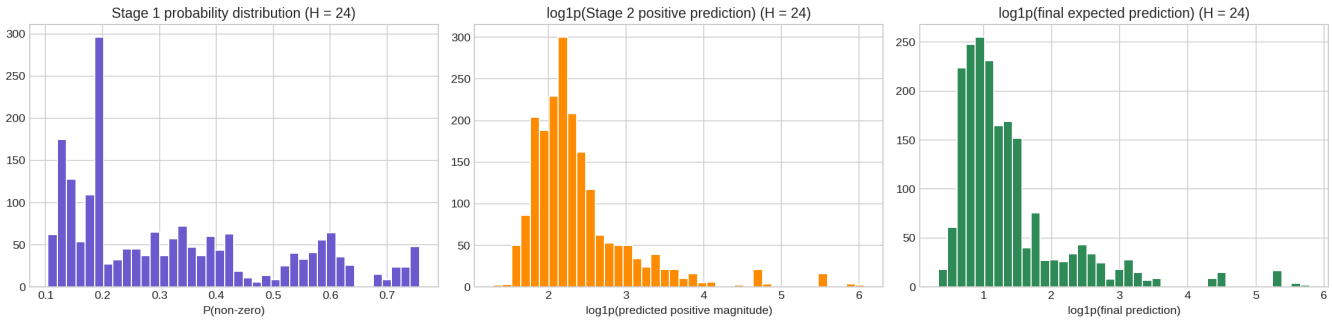
best_pair_confusion_h24.png

Best Pair Confusion 48H



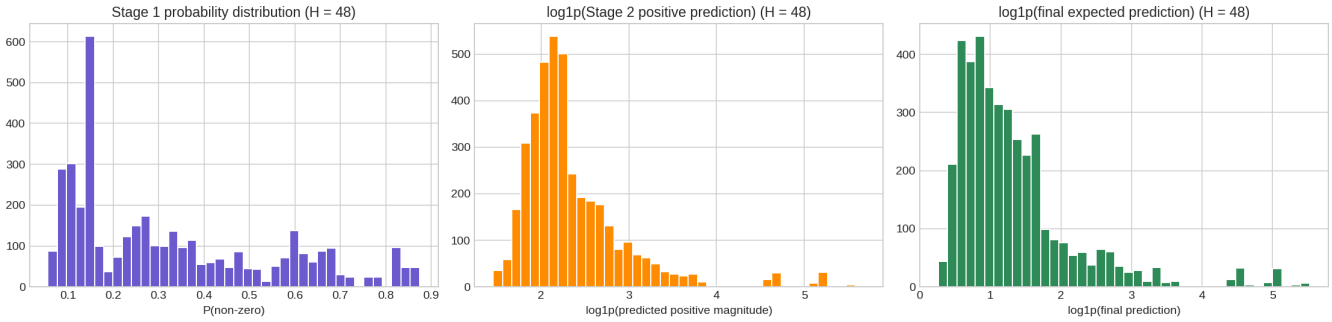
best_pair_confusion_h48.png

Prediction Distributions 24H



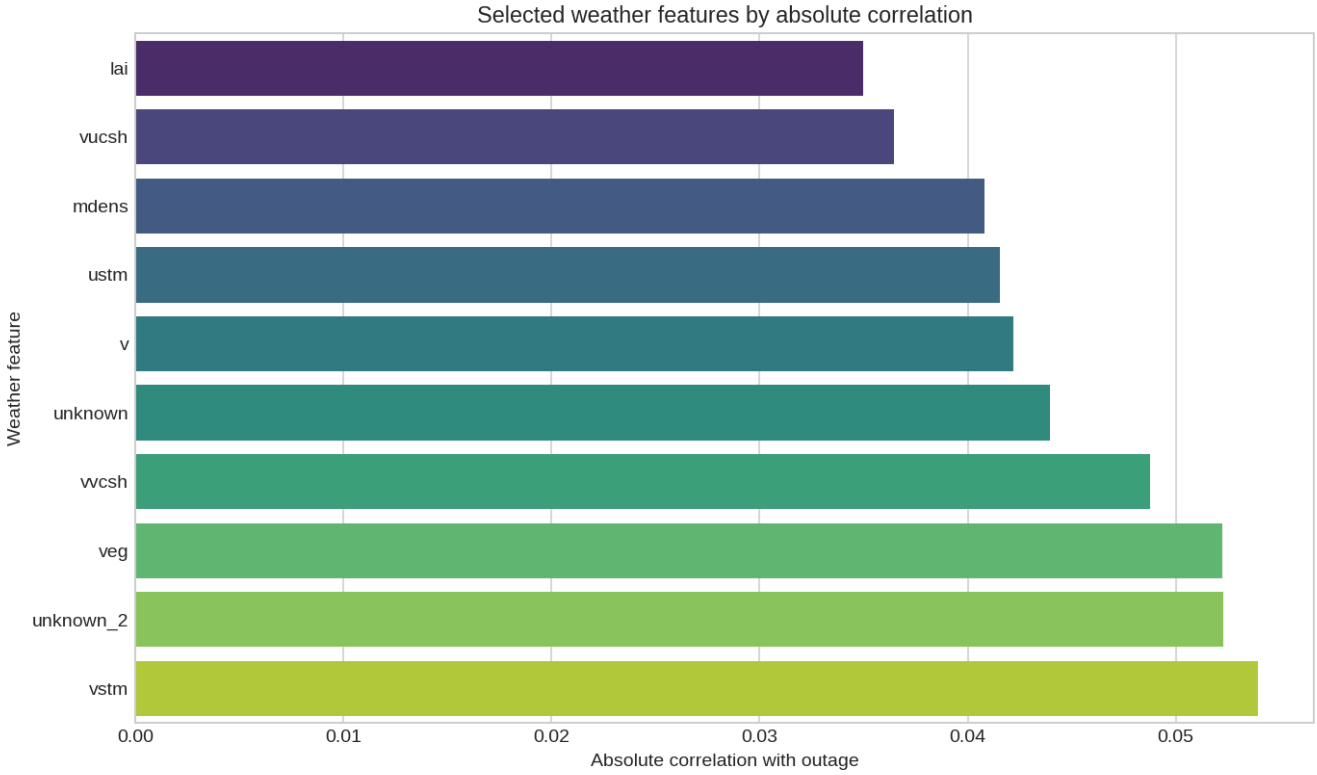
prediction_distributions_h24.png

Prediction Distributions 48H



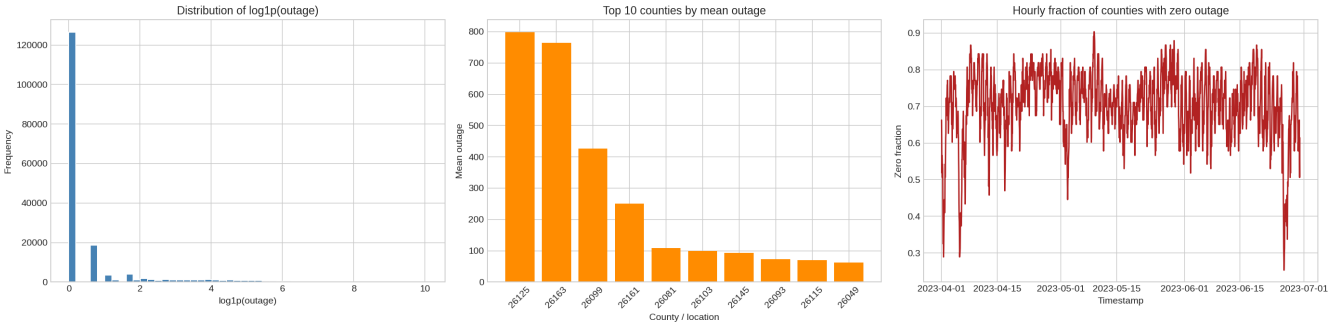
prediction_distributions_h48.png

Selected Weather Features



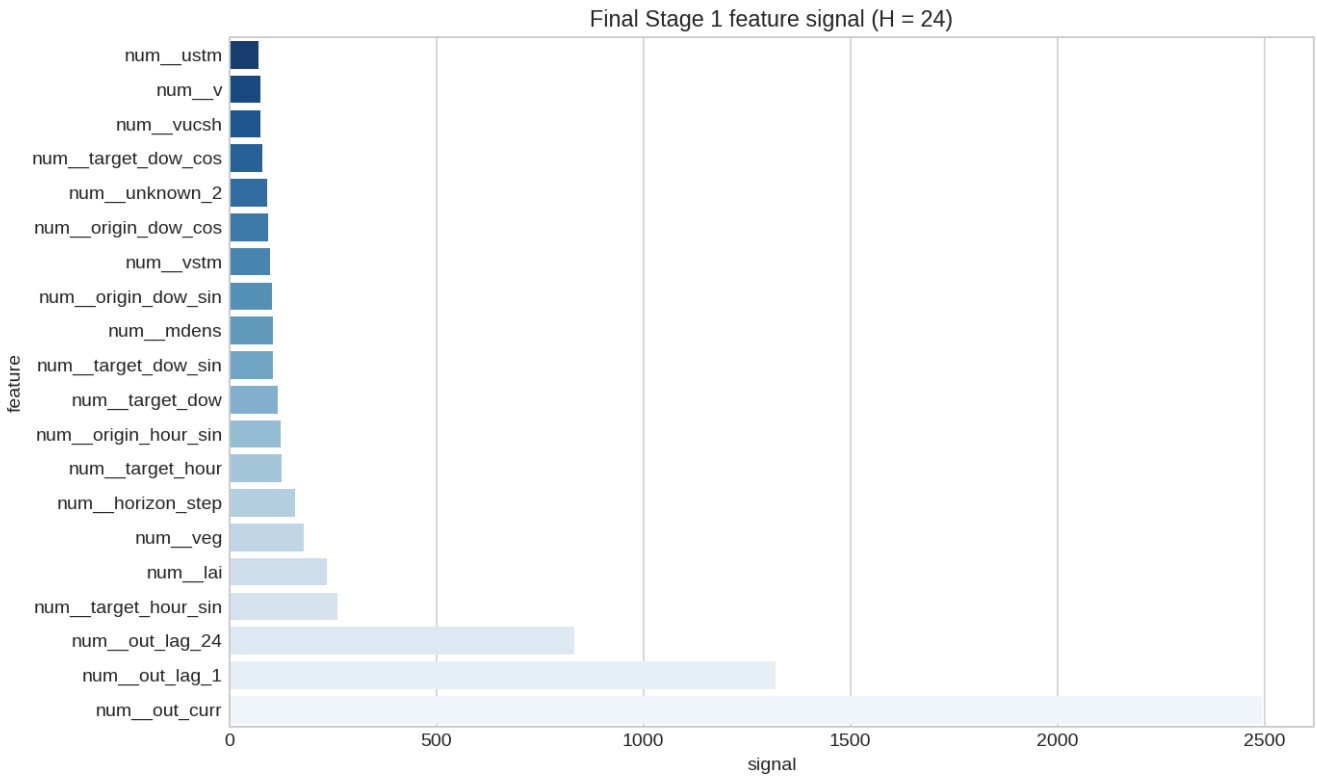
selected_weather_features.png

Eda Overview



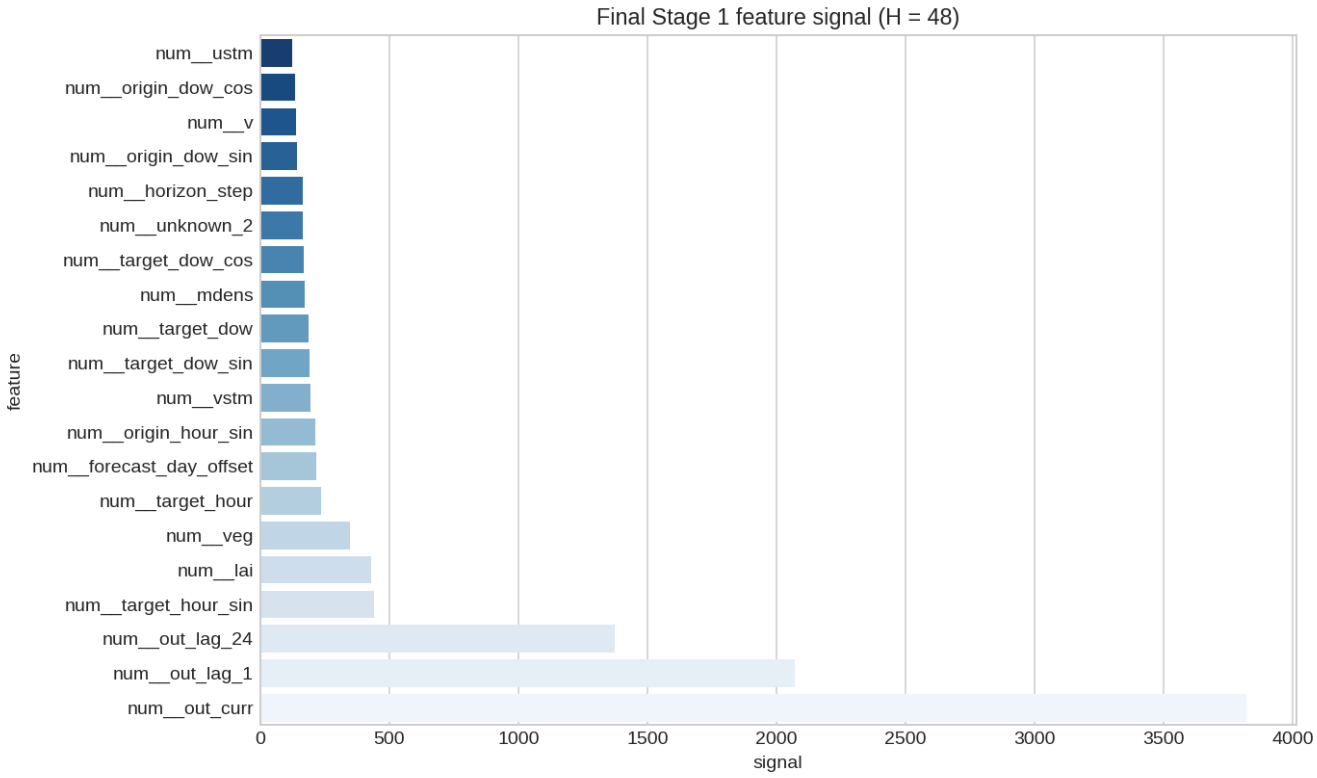
eda_overview.png

Final Stage1 Feature Signal 24H



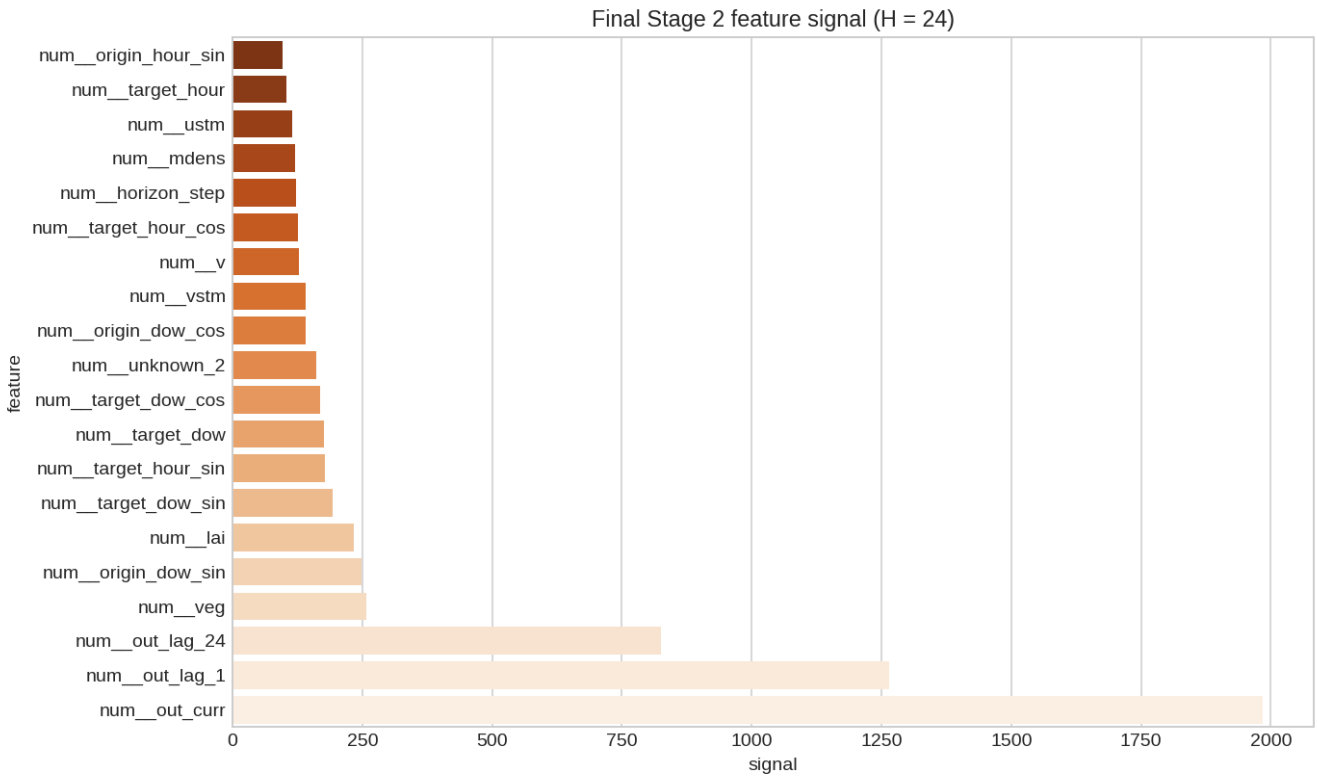
final_stage1_feature_signal_h24.png

Final Stage1 Feature Signal 48H



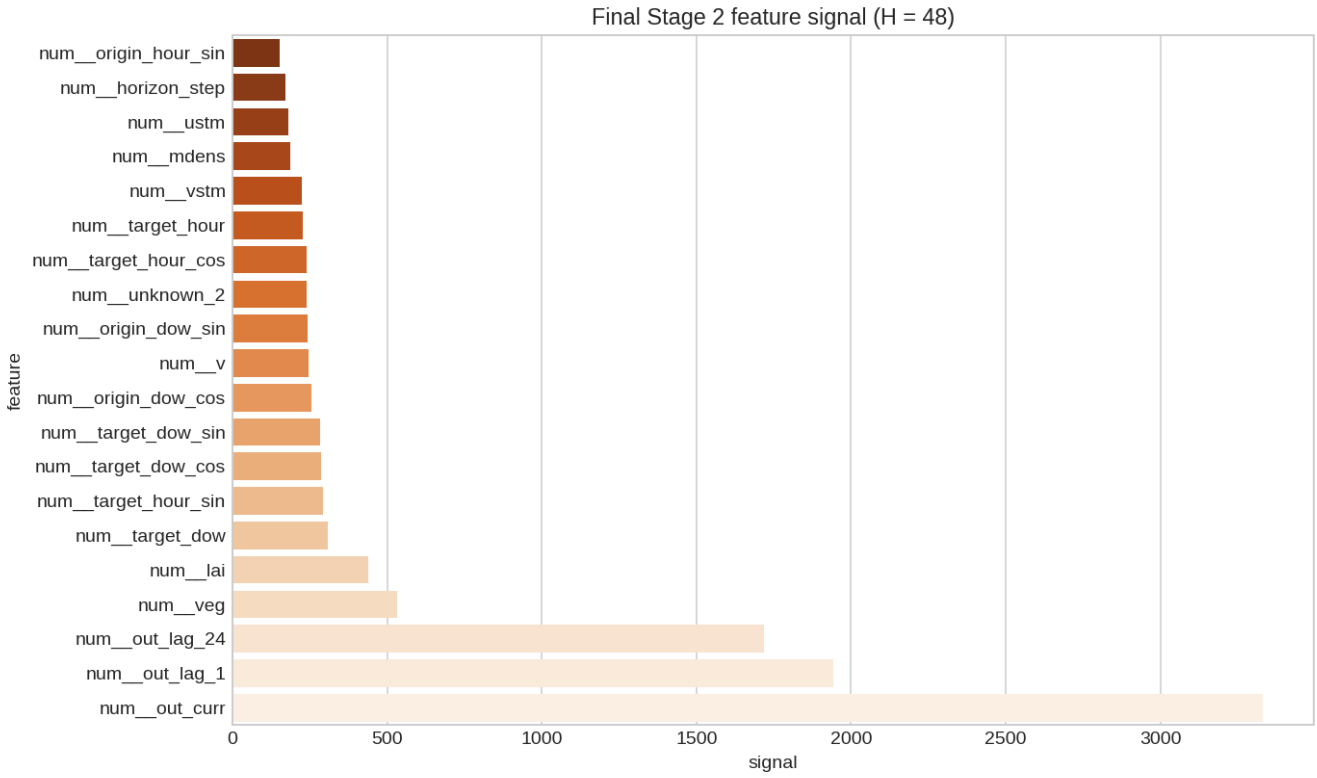
final_stage1_feature_signal_h48.png

Final Stage2 Feature Signal 24H



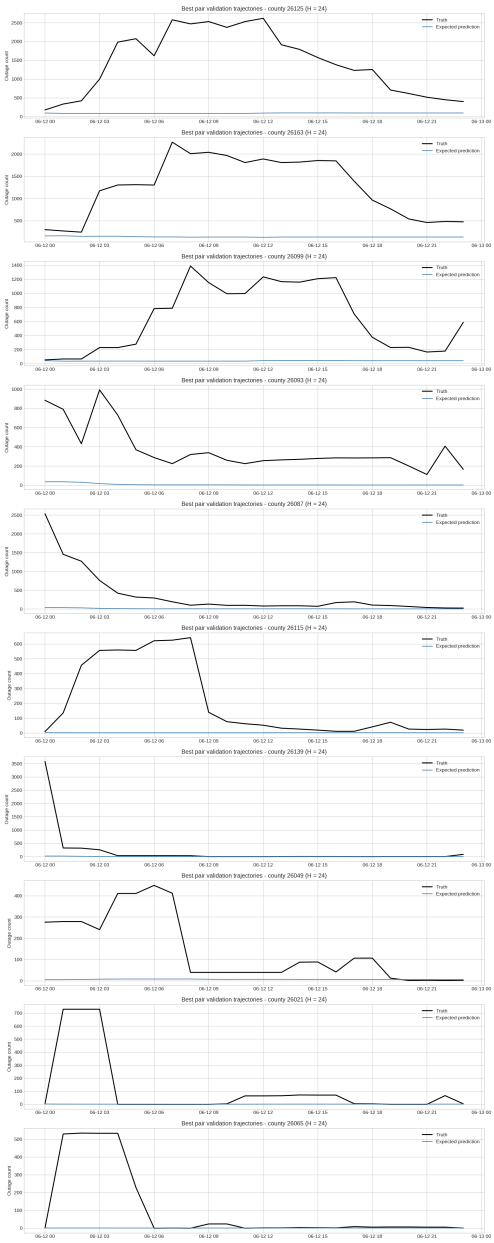
final_stage2_feature_signal_h24.png

Final Stage2 Feature Signal 48H



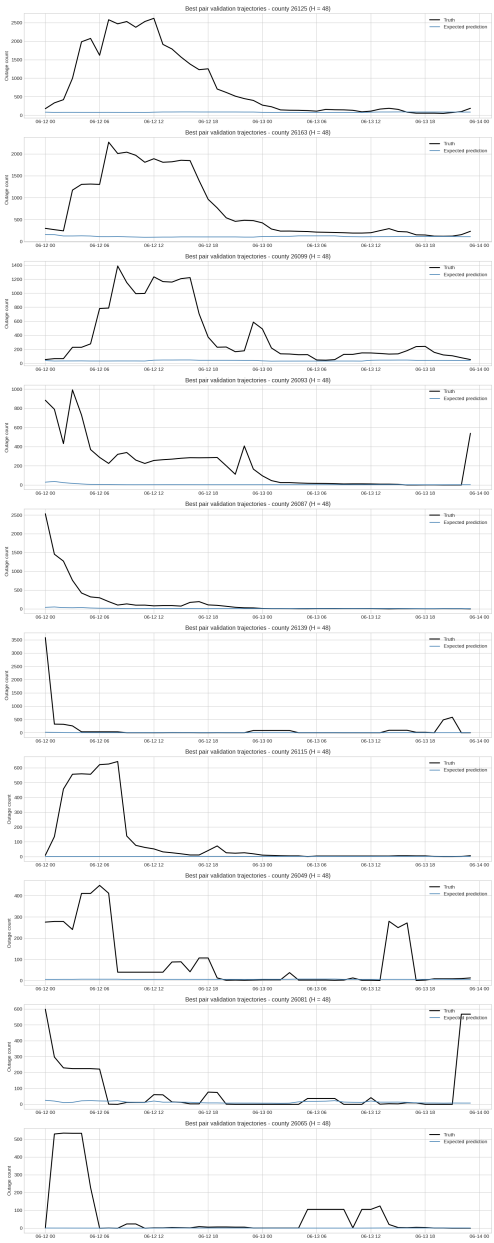
final_stage2_feature_signal_h48.png

Best Pair Validation Trajectories 24H



best_pair_validation_trajectories_h24.png

Best Pair Validation Trajectories 48H



best_pair_validation_trajectories_h48.png

Folder guide

- tables/: aggregated summaries, rankings, split definitions, and run-level CSV records.
- predictions/: exported submission-style predictions and detailed candidate validation predictions.
- models/: reusable .joblib artifacts plus metadata JSON files for submission and refit bundles.
- metrics/: fold-level metrics, training histories, and optional demo evaluation JSON files.
- plots/: visual diagnostics such as split layouts, model comparisons, confusion matrices, and feature-signal plots.
- checkpoints/: runtime resume artifacts used during notebook execution. This folder may be intentionally absent from GitHub uploads.